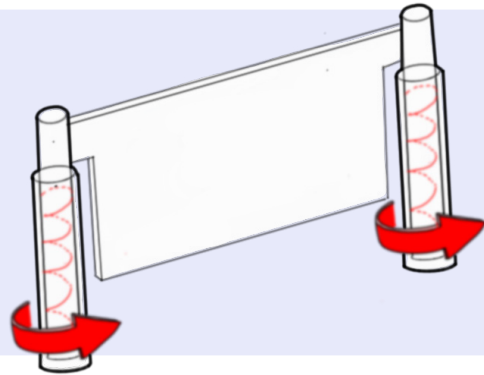


technical development

raising and locking system



screw threaded

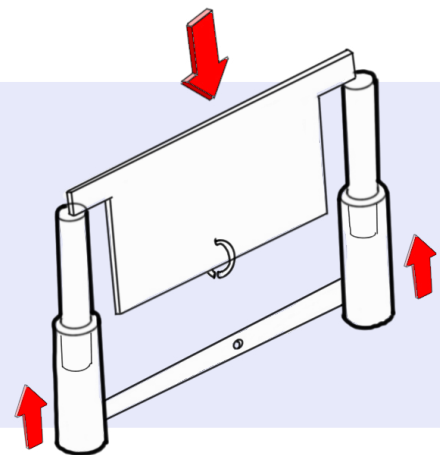
- complete control over height of board
- screw thread pitch can be adjusted to change rate of height increase



user testing and evaluation

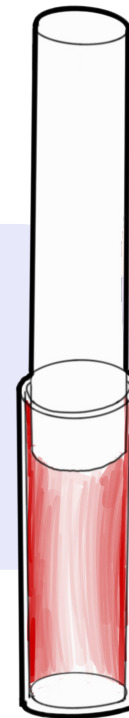


- screw motion is inconvenient and takes time
- may be hard to complete when weight of whiteboard is attached
- hard to turn both rails in sync



pneumatic cylinder

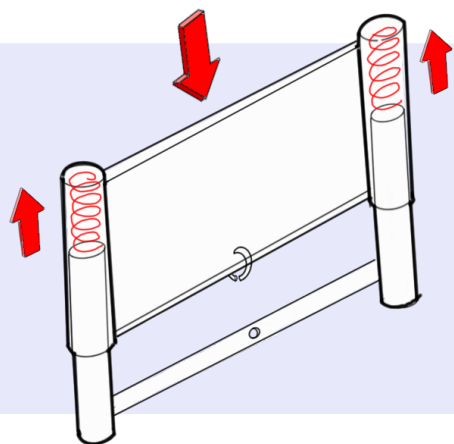
- holds the whiteboard in upwards position using air pressure
- user presses it down
- clips onto bar on the bottom



user testing and evaluation

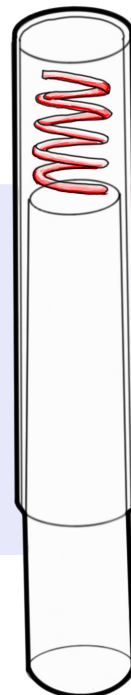


- very ergonomic raising of board, slow but firm
- air pressure needs to be decided as the cylinder that was ordered was far too powerful



spring loaded

- holds the whiteboard in upwards position
- user presses it down
- clips onto bar on the bottom



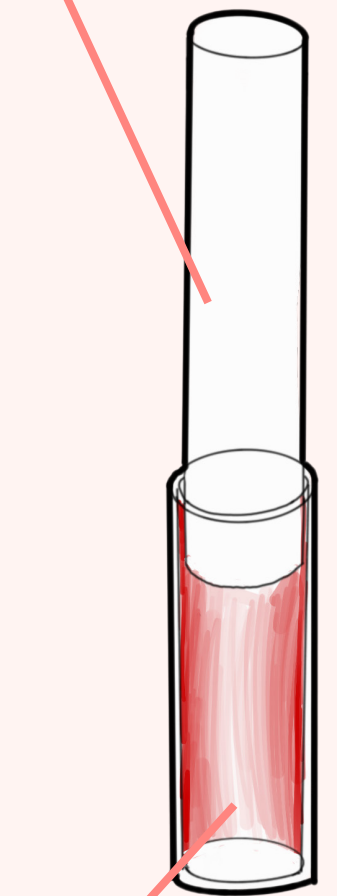
user testing and evaluation

- may be hard to control strength of spring and upwards force
- upon trying to make a prototype, it was quickly identified that this mechanism is not very feasible at all, due to the inavailability of springs at such a large size, and the difficulty of actually making the mechanism given constraints.

decision pneumatic cylinder

- This method seems by far the most feasible, whilst also prioritising the human experience with a gradual raising mechanism.

constant upwards force



high air pressure